Consider the lateral axis equations corresponding to the cruise condition of the L-1011 aircraft (pronounced "el-ten-eleven" and also known as Tristar). The state vector and the input vector are defined by

$$x(t) = \begin{bmatrix} x_1(t) & bank\ angle\ (rad) \\ x_2(t) & roll\ rate\ (rad/s) \\ x_3(t) & yaw\ rate\ (rad/s) \\ x_4(t) & sideslip\ angle\ (rad) \end{bmatrix}$$

$$u(t) = \begin{bmatrix} \delta_r & rudder\ deflection \\ \delta_a & aileron\ deflection \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1.89 & 0.39 & -5.555 \\ 0 & -0.034 & -2.98 & 2.43 \\ 0.034 & -0.0011 & -0.99 & -0.21 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 0.36 & -1.6 \\ -0.95 & -0.032 \\ 0.03 & 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x$$

Recall that the eigenvalues of the system are given by

$-1.4811 \pm 0.62915\, i$ $\quad(dutch\ roll\ mode)$

$-2.016157\ (roll\ subsidence\ mode)$

-0.1016 (spiral mode)

a) Design a static state-feedback controller $u = Kx$ such that the eigenvalues are as follows

   $-2.02506 + 2.02094\, i\ (dutch\ roll\ mode)$
   $-3.4146 \pm 2.96854\, i\ (roll\ mode)$

b) Choose $K = \begin{bmatrix} 0 & 0 & k_{13} & k_{14} \\ k_{21} & k_{22} & 0 & 0 \end{bmatrix}$ and assign the eigenvalues as given in a)

c) Design a full-order observer via Sylvester equation.

d) Using the results obtained in a) and c) (state-feedback via estimated states) design low order controllers, such that the tracking performance of the system is satisfactory for unit step reference signals.

e) Simulate the system and verify your results in Matlab or Mathematica.